

Fundamentals of Data Science - Lab Record

9. Property Data Analysis Using Pandas DataFrame Operations

Program:

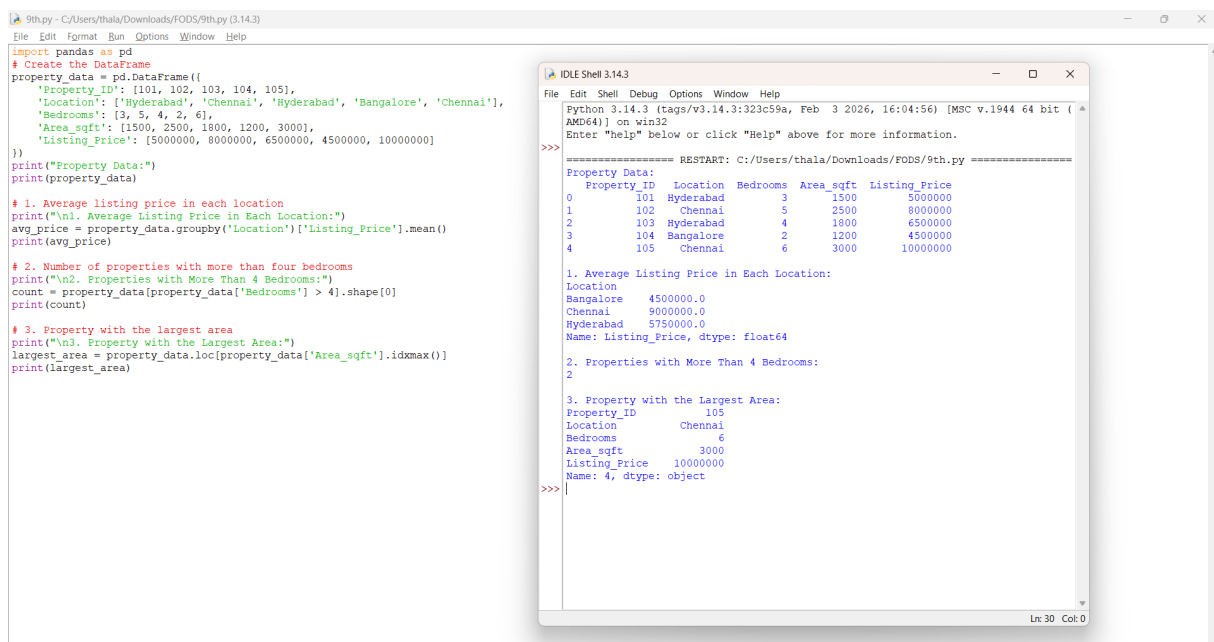
```
import pandas as pd
# Create the DataFrame
property_data = pd.DataFrame({
    'Property_ID': [101, 102, 103, 104, 105],
    'Location': ['Hyderabad', 'Chennai', 'Hyderabad', 'Bangalore', 'Chennai'],
    'Bedrooms': [3, 5, 4, 2, 6],
    'Area_sqft': [1500, 2500, 1800, 1200, 3000],
    'Listing_Price': [5000000, 8000000, 6500000, 4500000, 10000000]
})
print("Property Data:")
print(property_data)

# 1. Average listing price in each location
print("\n1. Average Listing Price in Each Location:")
avg_price = property_data.groupby('Location')['Listing_Price'].mean()
print(avg_price)

# 2. Number of properties with more than four bedrooms
print("\n2. Properties with More Than 4 Bedrooms:")
count = property_data[property_data['Bedrooms'] > 4].shape[0]
print(count)

# 3. Property with the largest area
print("\n3. Property with the Largest Area:")
largest_area = property_data.loc[property_data['Area_sqft'].idxmax()]
print(largest_area)
```

Output:



```
Python 3.14.3 (tags/v3.14.3:323c59a, Feb 3 2026, 16:04:56) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.

>>>
===== RESTART: C:/Users/thala/downloads/FODS/9th.py =====
Property Data:
  Property_ID  Location  Bedrooms  Area_sqft  Listing_Price
0         101  Hyderabad         3        1500         5000000
1         102    Chennai         5        2500         8000000
2         103  Hyderabad         4        1800         6500000
3         104  Bangalore         2        1200         4500000
4         105    Chennai         6        3000        10000000

1. Average Listing Price in Each Location:
Location
Bangalore    4500000.0
Chennai      9600000.0
Hyderabad    5750000.0
Name: Listing_Price, dtype: float64

2. Properties with More Than 4 Bedrooms:
2

3. Property with the Largest Area:
Property_ID      105
Location         Chennai
Bedrooms         6
Area_sqft       3000
Listing_Price    10000000
Name: 4, dtype: object
>>>
```

10. Data Visualization of Monthly Sales Using Matplotlib (Line Plot and Bar Plot)

Program:

```
import matplotlib.pyplot as plt

# Monthly Sales Data
months = ["Jan", "Feb", "Mar", "Apr", "May", "Jun"]
sales = [25000, 30000, 28000, 35000, 40000, 45000]

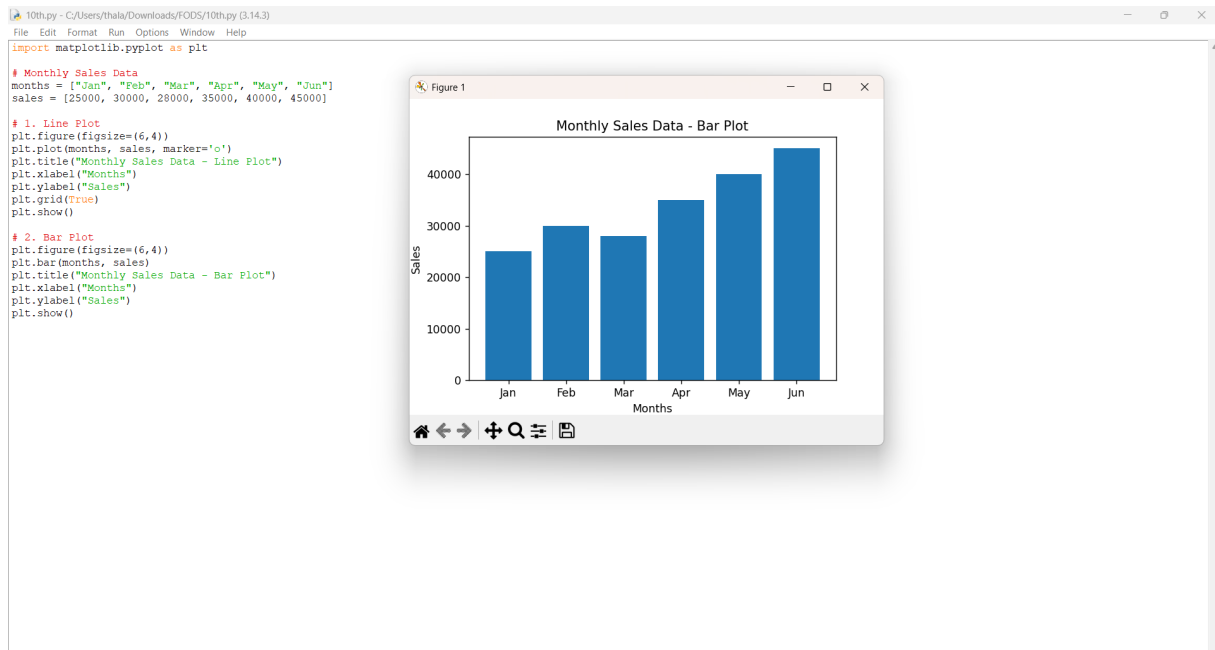
# 1. Line Plot
plt.figure(figsize=(6,4))
plt.plot(months, sales, marker='o')
plt.title("Monthly Sales Data - Line Plot")
plt.xlabel("Months")
plt.ylabel("Sales")
plt.grid(True)
plt.show()

# 2. Bar Plot
plt.figure(figsize=(6,4))
plt.bar(months, sales)
plt.title("Monthly Sales Data - Bar Plot")
plt.xlabel("Months")
plt.ylabel("Sales")
plt.show()
```

Output - Line Plot:



Output - Bar Plot:



11. Data Visualization of Monthly Product Sales Using Matplotlib (Line Plot, Scatter Plot, and Bar Plot)

Program:

```
import matplotlib.pyplot as plt

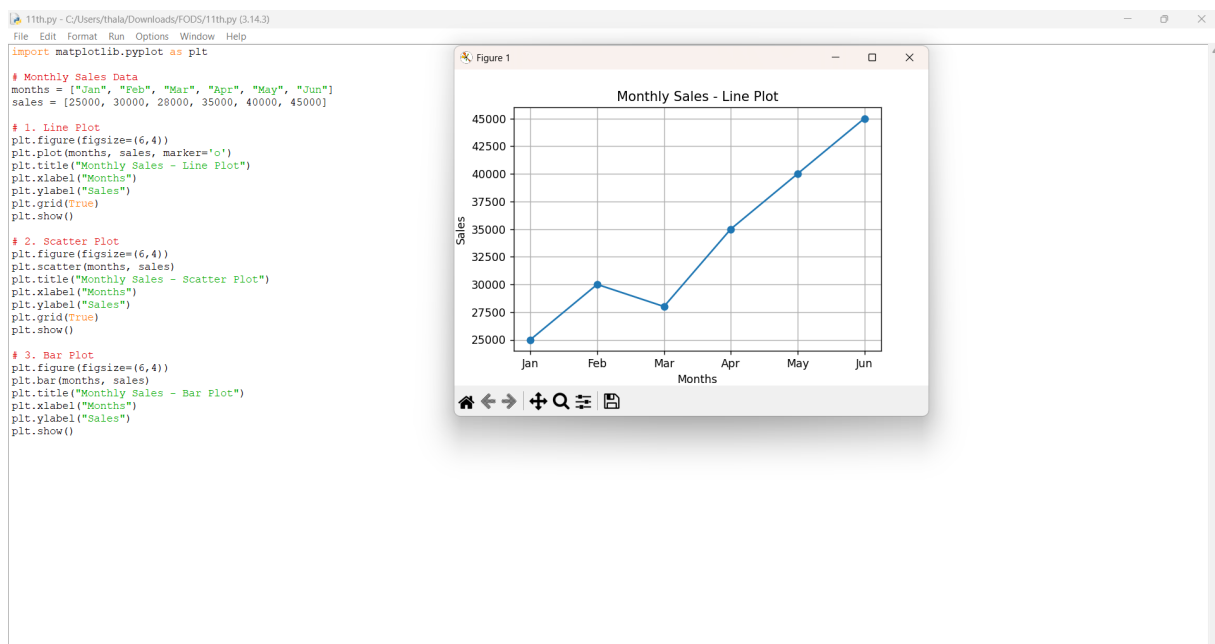
# Monthly Sales Data
months = ["Jan", "Feb", "Mar", "Apr", "May", "Jun"]
sales = [25000, 30000, 28000, 35000, 40000, 45000]

# 1. Line Plot
plt.figure(figsize=(6,4))
plt.plot(months, sales, marker='o')
plt.title("Monthly Sales - Line Plot")
plt.xlabel("Months")
plt.ylabel("Sales")
plt.grid(True)
plt.show()

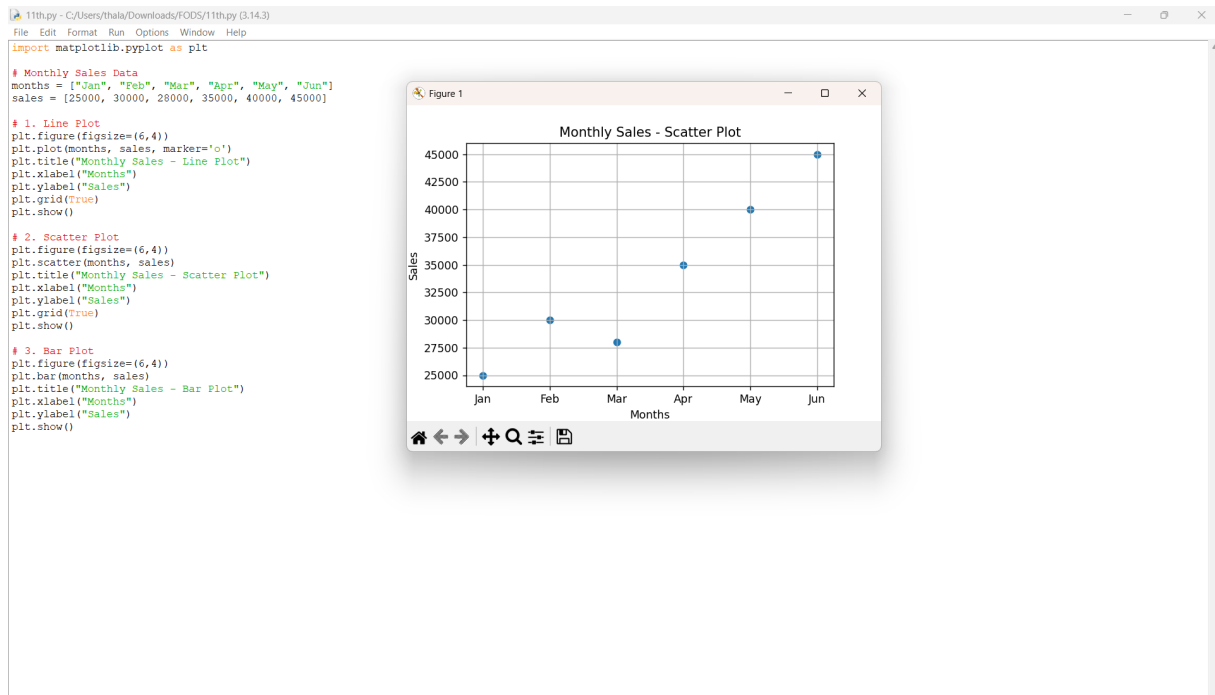
# 2. Scatter Plot
plt.figure(figsize=(6,4))
plt.scatter(months, sales)
plt.title("Monthly Sales - Scatter Plot")
plt.xlabel("Months")
plt.ylabel("Sales")
plt.grid(True)
plt.show()

# 3. Bar Plot
plt.figure(figsize=(6,4))
plt.bar(months, sales)
plt.title("Monthly Sales - Bar Plot")
plt.xlabel("Months")
plt.ylabel("Sales")
plt.show()
```

Output - Line Plot:



Output - Scatter Plot:



Output - Bar Plot:

